

29-07-26

Tracing of quicksort of 54, 12, 19, 3, 8, 46, 31
 $qs(a, 0, 6)$

SOL:- first = 0, last = 6 pivot = 0

54	12	19	3	8	46	31
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i / i / / / / j

i	j	a[i]	a[j]	a[pivot]
0	6	54	31	54 $54 < 54$
1	6	12	31	54 $12 < 54$
2	6	19	31	54 $19 < 54$
3	6	3	31	54 $3 < 54$
4	6	8	31	54 $8 < 54$
5	6	46	31	54 $46 < 54$

swap(a[pivot], a[i])

So,

3	12	19	8	31	46	54
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3	12	19	8	31	46	54
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i / j /
j /

$i > j \Rightarrow$

swap pivot & j

3	12	19	8	31	46	54
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3	12	6	19	31	46	54
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3 8 12 19 31 46 54

Q.

3 8 12 19 31 46 54

i	j	a[i]	a[j]	a[pivot]
0	5	31	46	31
3	4	3	8	31
0	5	31	46	31
5	4	46	3	31
0	3	3	8	3
1	0	12	3	3
1	3	12	8	12
2	3	19	8	12
1	3	12	19	12
3	2	19	8	12

initial

$i < j \Rightarrow \text{swap}$

$i > j \Rightarrow \text{swap pivot}$

initial

$i > j \Rightarrow \text{swap } j \text{ \& pivot}$

initial

swap $j \text{ \& } i$

initial

$i > j \Rightarrow \text{swap pivot}$

∴ final array = 3 1 12 19 31 46 54